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# =====
# Image Sharpening using Laplacian Mask
# Positive Center Coefficient (8-Neighbour)
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import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload Image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Positive Center Laplacian Kernel
kernel = np.array([
    [-1, -1, -1],
    [-1,  8, -1],
    [-1, -1, -1]
], dtype=np.float32)

# Apply Laplacian filter
laplacian = cv2.filter2D(img_rgb, cv2.CV_32F, kernel)

# Convert original image to float
img_float = img_rgb.astype(np.float32)

# Sharpen Image
sharpened = img_float + laplacian

# Clip pixel values
sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)

# Display Results
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(np.clip(np.abs(laplacian),0,255).astype(np.uint8))
plt.title("Laplacian Output")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(sharpened)
plt.title("Sharpened Image")
plt.axis("off")

plt.show()

# Save Output
cv2.imwrite("Sharpened_Positive_Laplacian.jpg",
           cv2.cvtColor(sharpened, cv2.COLOR_RGB2BGR))

print("Output saved as Sharpened_Positive_Laplacian.jpg")

files.download("Sharpened_Positive_Laplacian.jpg")

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Original Image



Laplacian Output



Sharpened Image



Output saved as Sharpened_Positive_Laplacian.jpg